

An Investigation of Knowledge-Based AI vs. Human Evaluation in the Context of Academic Summary Evaluation: Similarities, Dissimilarities, and Being Toward Mutual Understandings

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Abstract: This study aims to explore the similarities and dissimilarities of knowledge-based AI evaluations vs. human evaluations and discuss how they can be utilized for formative feedback in academic summary writing. Data were collected from 62 students who utilized AI-based formative feedback to make revisions to their summaries. We compared indices on three dimensions (surface, structure, semantic) that were automatically generated through this software with human-rated evaluations. MANOVA results of learners' initial draft and final revision showed learning gains in the semantic dimension and human-evaluated scores. Some significant correlations were observed between automatic and human-rated evaluations. Given that each measure can be interpreted to provide different insights, we suggest combining knowledge-based AI and human evaluations for rich and informative feedback.

Introduction

Instructors have used summary writing to assess reading comprehension and to scaffold students' concept learning (Dunlosky et al., 2013; Kim & McCarthy, 2021). Namely, learners can acquire knowledge while processing the given material, understanding its structures, identifying key concepts, and organizing these into words. To support this process, evaluating learner performance and providing timely feedback is important.

A typical means to evaluate summaries of texts is either through knowledge-based AI systems or human evaluation. The use of knowledge-based AI for evaluation is an emerging field, fueled by the development of natural language processing techniques (Hu et al., 2022). Technology-enabled assessment is perceived as objective, free from bias and stereotypes, able to produce timely feedback to learners, and having potential when there is a lack of human evaluators (Belz & Reiter, 2006; Lewis Sevcikova, 2018; Mithun et al., 2012). However, with many techniques still focused mainly on corpus-based metrics, key concepts have to be identified beforehand and only mathematically calculatable variables are able to be judged (Belz & Reiter, 2006; Mithun et al., 2012). Thus, there have been arguments that automatic evaluations are yet to replace human evaluations (Hu et al., 2022). On the other hand, human evaluation is able to judge summaries holistically, taking argument qualities and aesthetic styles into account (Lewis Sevcikova, 2018). However, human ratings consume much time and effort, and training evaluators is also a challenging task (Brookhart & Chen, 2015; Lewis Sevcikova, 2018). There is also the issue of human raters showing low correlations between each other, especially when no clear rubric exists to evaluate (Belz & Reiter, 2006; Hu et al., 2022; Mithun et al., 2012).

As such, the characteristics of these methods of evaluations differ in many ways. In this study, we aim to use academic summaries to explore the relationship between the two evaluation indices and discuss how they can supplement each other.

Methods

Research context and data source

Data were collected from 62 students across six semesters of an instructional technology course. The course was offered in an asynchronous online format, with learners participating in weekly summary writing, quizzes, and discussions. While learners submitted their summaries directly to the university learning management system for most of the semester, they used a web-based technology that provides knowledge-based AI evaluation and formative feedback for one week. The system provided learners with multimodal feedback, including visual and textual information, and learners were allowed to make as many revisions as they preferred. We used the summaries and indices generated from this interaction for this research.

Knowledge-based AI evaluation

We used the Student Mental Model Analyzer for Research and Teaching (SMART) system to compute knowledge-based AI evaluation indices. SMART is an automated summary evaluator which provides formative

feedback on submitted summary by utilizing concept maps as re-represented mental models. Based on graph theory (Wasserman & Faust, 1994), SMART generates concept map parameters for surface, structural, and semantic dimensions (3S). These measures are then compared to a reference model (an expert model) and 11 similarity measures are calculated. The resulting indices are shown in Table 1.

Table 1
SMART-Based Evaluation Indices

Dimension	Similarity Measures	Definition
Surface	S-Number of Concepts	The total number of concepts (nodes)
	S-Number of Relations	The total number of concept relations (links)
	S-Density	The cohesiveness of a concept map
Structure	S-Average Degree	The average number of degrees
	S-Mean Distance	The closeness of concepts (average geodesic distance)
	S-Diameter	Understanding broadness (largest geodesic distance)
Semantic	Concept Matching	Semantically identical concepts
	Propositional Matching	Fully identical propositions
	Balanced Semantic Matching	Propositional matching divided by concept matching
	Recall-C	The proportion of key concepts
	Recall-P	The proportion of key relations

Note. Modified from Kim & McCarthy (2021, p.691).

Human evaluation

Human evaluation was conducted by two researchers based on the rubric introduced by Gao et al. (2019). Each summary was scored on a scale from zero to five in terms of their content quality, content coverage, content coherence, and argument quality. Cronbach's Alpha was used to examine the interrater reliability of the five indices. Content quality ($\alpha = .88$), content coverage ($\alpha = .83$), content coherence ($\alpha = .74$), and argument quality ($\alpha = .90$) showed sufficient levels of reliability (Graham, 2012).

Data analysis

To explore if there are gains between learners' initial and final summary submissions, we conducted one-way repeated-measures multivariate analysis of variance (MANOVAs). A total of four MANOVAs were used to examine the three dimensions of the SMART-generated similarity measures as well as the human evaluation. The relation between the two types of indices were explored using Pearson's coefficient of correlation.

Findings

We conducted a series of MANOVAs to investigate if revisions could lead learners' summaries to become similar to the reference (Table 2). Surface level indices showed some improvement in terms of descriptive data but were not statistically significant (Wilks' Lambda = 0.95, $p = 0.35$). This could have been due to the assignment having clear guidelines about the length of the summaries, leaving little room for change. There was also no significant difference for the structural measures (Wilks' Lambda = 0.95, $p = 0.35$). As with surface measures, having a clear restriction on the number of words could have limited structural changes from taking place. In terms of the semantic dimension, there was a significant difference (Wilks' Lambda = 0.32, $p < 0.001$). Univariate repeated-measures ANOVAs showed statistically significant gains for all semantic measures. Learners' revisions were focused on the semantic area, showing clear changes in the use of key concepts and relations. As SMART provides feedback focusing on terms and relations that should be included in summaries, learners seem to have been able to add such information through revisions. Human-rated indices also showed significant gains (Wilks' Lambda = 0.52, $p < 0.001$), with univariate repeated-measures ANOVAs confirming that these were present in all four indices. The findings suggest that from a rubric-based human view, learners performed better through revisions.

Table 2
Similarity Scores and Human Rated Descriptors: Means, Standard Deviations, MANOVA Results

Descriptor	Mean (SD)		(F – I)	F	Sig.	η^2
	Initial (I)	Final (F)				
<i>Surface</i>						
S-Number of Concepts	0.76 (0.14)	0.78 (0.15)	0.02	2.25	0.138	0.40
S-Number of Relations	0.72 (0.15)	0.75 (0.16)	0.03	2.94	0.092	0.50
S-Density	0.77 (0.14)	0.79 (0.15)	0.02	1.90	0.173	0.30

Structure

S-Average Degree	0.95 (0.04)	0.95 (0.04)	0.007	3.27	0.075	0.50
S-Mean Distance	0.91 (0.07)	0.91 (0.06)	<0.001	0.01	0.916	<0.01
S-Diameter	0.84 (0.12)	0.84 (0.10)	0.005	0.08	0.779	<0.01

Semantic

Concept Matching	0.26 (0.12)	0.36 (0.13)	0.10	69.59***	<.001	0.53
Propositional Matching	0.06 (0.05)	0.11 (0.08)	0.05	44.29***	<.001	0.42
Balanced Semantic Matching	0.21 (0.13)	0.28 (0.13)	0.08	26.08***	<.001	0.30
Recall-C	0.42 (0.16)	0.70 (0.22)	0.28	120.45***	<.001	0.66
Recall-P	0.29 (0.16)	0.61 (0.28)	0.33	113.68***	<.001	0.65

Content Quality	2.73 (0.58)	3.12 (0.61)	0.40	35.64***	<.001	0.37
Content Coverage	2.73 (0.58)	3.21 (0.56)	0.48	47.91***	<.001	0.44
Content Coherence	1.84 (0.49)	2.06 (0.59)	0.23	17.79***	<.001	0.23
Argument Quality	2.74 (0.56)	3.10 (0.62)	0.35	32.41***	<.001	0.35

Note. $N = 62$. *** $p \leq .001$

Pearson's correlation coefficients were calculated between the SMART-generated 3S similarity measures and human evaluations to explore their relationships (Table 3). For the initial summary scores, a statistically significant correlation was observed between Recall-P and content quality, Recall-C and content coverage, and Recall-P and content coverage ($r(60) = .25, .27$, and $.30$ with $p = .05, .04$, and $.02$, respectively). In the case of final summary scores, we observed significant values with structure and semantic indices. Specifically, the mean distance similarity values showed correlations with content coverage and content coherence ($r(60) = .32$ and $.26$ with $p = .01$ and $.05$, respectively). For semantic indices, concept matching, propositional matching, balanced semantic matching, and recall-C ($r(60) = .35, .42, .43$, and $.25$ with $p = .005, .001, <.001$ and $.05$, respectively) showed relations with content quality. There were also significant correlations between balanced semantic matching and content coherence, propositional matching and argument quality, and balanced semantic matching and argument quality ($r(60) = .27, .29$, and $.30$ with $p = .03, .02$, and $.02$, respectively).

In summary, some significance was identified, with a few within the range of previous studies on such relations (Jonsson & Svingby, 2007). While initial submission evaluations had few significant values, the final scores showed more semantic indices showing significance. Such differences could have stemmed from the characteristics of automatic assessment. SMART's indices measure similarities with a more micro view, focusing on if the certain concept or relation is explicitly stated within the text. On the other hand, human evaluators can have a more comprehensive view and are able to understand underlying meanings. Thus, this could have caused a discrepancy between SMART and human rater's judgements in initial summaries. As learners utilized SMART's feedback for revision, they could have been encouraged to identify more key terms and to state them clearly. This could have led to the higher levels of correlation between SMART indices and human evaluations and would be in line with the explanations given by Dang & Owczarzak (2008), who stated automatic evaluations could lead to low scores when differently phrased information is not correctly recognized.

Table 3
Correlation between Similarity Scores and Human Evaluations

Descriptor	Initial				Final			
	1	2	3	4	1	2	3	4
<i>Surface</i>								
S-Number of Concepts	-0.14	-0.07	-0.03	0.03	-0.09	-0.11	-0.02	-0.08
S-Number of Relations	-0.06	-0.01	0.03	0.08	-0.04	-0.1	0.03	-0.08
S-Density	-0.12	-0.03	-0.01	0.08	-0.12	-0.1	-0.04	-0.08
<i>Structure</i>								
S-Average Degree	0.02	0.01	0.00	-0.05	0.006	-0.12	0.08	-0.08
S-Mean Distance	0.01	0.11	0.04	0.13	0.16	0.32**	0.26*	0.20
S-Diameter	-0.16	-0.01	-0.02	0.03	0.02	0.16	0.11	-0.02
<i>Semantic</i>								
Concept Matching	0.07	0.17	0.06	0.12	0.35**	0.21	0.16	0.21
Propositional Matching	0.07	0.21	-0.05	0.08	0.42***	0.23	0.21	0.29*

Balanced Semantic Matching	0.06	0.21	-0.01	0.06	0.43***	0.23	0.27*	0.30*
Recall-C	0.22	0.27*	-0.01	0.14	0.25*	0.19	0.00	0.12
Recall-P	0.25*	0.30*	-0.04	0.16	0.25	0.18	0.03	0.11

Note. $N = 62$. * $p \leq .05$. ** $p \leq .01$. *** $p \leq .001$ 1=Content Quality, 2= Content Coverage, 3= Content Coherence, 4= Argument Quality.

Discussion and implications

This study aims to explore the relationship between knowledge-based AI and human in academic summary evaluation. To this end, we compared SMART-generated indices alongside rubric-based human evaluations. We found that system-generated evaluation is quite different from human judgements. While knowledge-based AI can be objective, focuses on micro-level changes, and looks for explicitly expressed information, human generated measures are more subjective, comprehensive, and holistic. As such, each measure needs to be interpreted accordingly. Furthermore, knowledge-based AI and human evaluations can supplement each other to provide insights the other cannot. Thus, we believe that using both measures collectively would allow for more informative formative feedback.

References

- Belz, A., & Reiter, E. (2006). Comparing automatic and human evaluation of NLG systems. *Conference of the European Chapter of the Association for Computational Linguistics*.
- Brookhart, S. M., & Chen, F. (2015). The quality and effectiveness of descriptive rubrics. *Educational Review*, 67(3), 343–368. <https://doi.org/10.1080/00131911.2014.929565>
- Dang, H. T., & Owczarzak, K. (2008). Overview of the TAC 2008 update summarization task. *Theory and Applications of Categories*.
- Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, 14(1), 4–58. <https://doi.org/10.1177/1529100612453266>
- Gao, Y., Driban, A., Xavier McManus, B., Musi, E., Davies, P., Muresan, S., & Passonneau, R. J. (2019). Rubric reliability and annotation of content and argument in source-based argument essays. *Proceedings of the Fourteenth Workshop on Innovative Use of NLP for Building Educational Applications*, 507–518. <https://doi.org/10.18653/v1/W19-4452>
- Graham, M., Milanowski, A., & Miller, J. (2012). Measuring and promoting inter-rater agreement of teacher and principal performance ratings. *Online Submission*.
- Hu, X., Chen, Q., Wang, H., Xia, X., Lo, D., & Zimmermann, T. (2022). Correlating automated and human evaluation of code documentation generation quality. *ACM Transactions on Software Engineering and Methodology*, 31(4), 1–28. <https://doi.org/10.1145/3502853>
- Jonsson, A., & Svingby, G. (2007). The use of scoring rubrics: Reliability, validity and educational consequences. *Educational Research Review*, 2(2), 130–144. <https://doi.org/10.1016/j.edurev.2007.05.002>
- Kim, M., & McCarthy, K. S. (2021). Improving summary writing through formative feedback in a technology-enhanced learning environment. *Journal of Computer Assisted Learning*, 37(3), 684–704. <https://doi.org/10.1111/jcal.12516>
- Lewis Sevcikova, B. (2018). Human versus automated essay scoring: A critical review. *Arab World English Journal*, 9(2), 157–174. <https://doi.org/10.24093/awej/vol9no2.11>
- Mithun, S., Kosseim, L., & Perera, P. (2012). Discrepancy between automatic and manual evaluation of summaries. *Proceedings of Workshop on Evaluation Metrics and System Comparison for Automatic Summarization*, 44–52.
- Wasserman, S., & Faust, K. (1994). *Social network analysis: Methods and applications*. Cambridge University Press.

Acknowledgments

The material is based upon work supported by the National Science Foundation under Grant No. 2247790 and Grant No. 2112532. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.